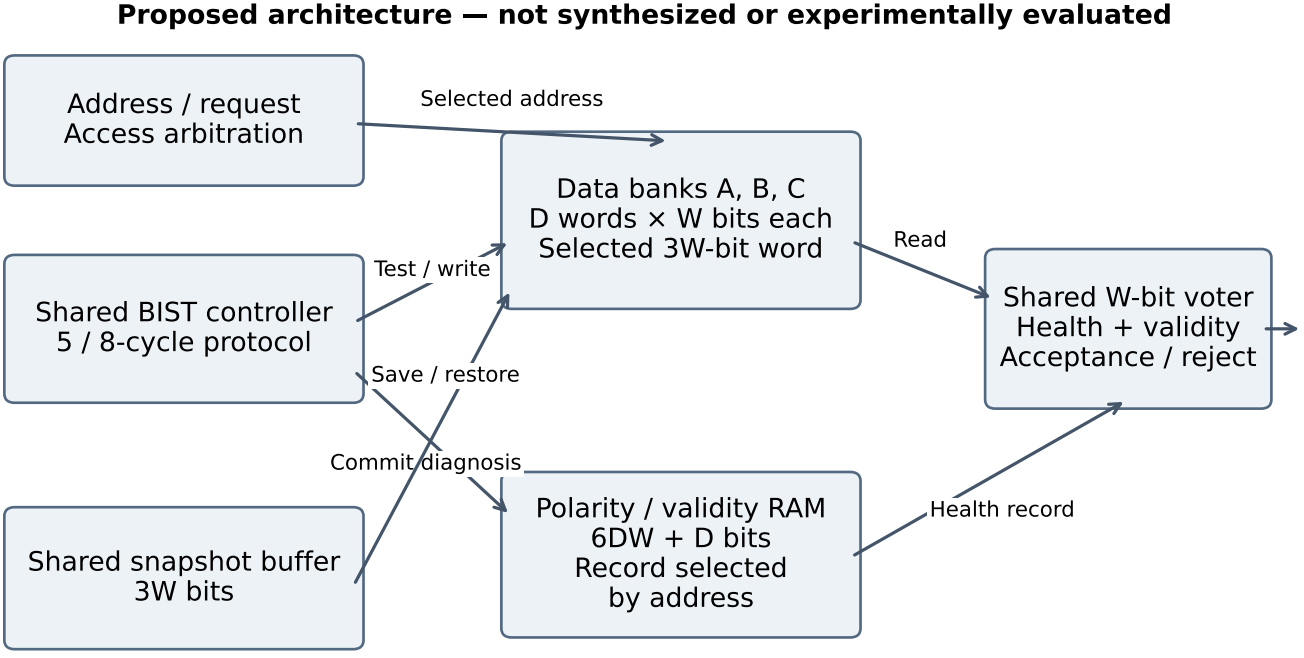
Proposed architecture — not synthesized or experimentally evaluated
Address / request
Access arbitration
Selected address
Data banks A, B, C
D words × W bits each
Selected 3W-bit word
Read
Shared W-bit voter
Health + validity
Acceptance / reject
Shared BIST controller
5 / 8-cycle protocol
Test / write
Save / restore
Commit diagnosis
Shared snapshot buffer
3W bits
Polarity / validity RAM
6DW + D bits
Record selected
by address
Health record